

4	(a) (i) diameter and extension: micrometer (screw gauge) or digital calipers	B1	
	length: tape measure or metre rule	B1	
	load: spring balance or Newton meter	B1	[3]
	(ii) to reduce the effect of random errors or to plot a graph to check for zero error in measurement of extension or to see if limit of proportionality is exceeded	B1	[1]
	(b) plot a graph of F against e and determine the gradient	B1	
	$E = (\text{gradient} \times l) / [\pi d^2 / 4]$	B1	[2]